

Rui Gong

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology
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RESEARCH INTERESTS Semidefinite Optimization and its application to Combinatorial Optimization, Convex Optimization and other areas.

EDUCATION

Georgia Institute of Technology Atlanta, Georgia, U.S.A.
PhD in Operation Research 2023.08-2027.08 (expected)
• Advisor: Diego Cifuentes, Alejandro Toriello
• Passed Comprehensive Exams

University of Waterloo Waterloo, Ontario, Canada
M.Math. in Combinatorics & Optimization 2021.09-2023.06
• Thesis: Low-Rank Plus Sparse Decompositions of Large-Scale Matrices via Semidefinite Optimization
• Advisor: Levent Tuncel
• Passed C&O PhD first stage comprehensive exams: Continuous Optimization, Graph Theory

University of Waterloo Waterloo, Ontario, Canada
B.Math. Triple Major in Mathematical Finance, Mathematical Optimization, Statistics 2017.09-2021.08
• Dean's Honour List

SCHOLARSHIPS & AWARDS

Georgia Institute of Technology
Ronald & Carol Beerman Fellowship 2024-2025
ISyE Premium Fellowship
Ronald & Carol Beerman Fellowship 2023-2024

University of Waterloo
Math Domestic Graduate Student Award 2022-2023
Sinclair Graduate Scholarship 2023.01-2023.04
Sinclair Graduate Scholarship 2022.09-2022.12
C & O Graduate Award 2022.01-2022.04
Sinclair Graduate Scholarship 2022.01-2022.04
President's Scholarship of Uwaterloo 2017

RESEARCH

Rounding the Lovász Theta Function with a Value Function Approximation (under major revision of SIAMOPT), Rui Gong, Diego Cifuentes, Alejandro Toriello.
Dynamic Interval Scheduling with Random Start and End Times (submitted), Rui Gong, Alejandro Toriello
Rounding the SDP Relaxations of the Quadratic Assignment Problem (in progress), Rui Gong, Alejandro Toriello

SEMINARS

Talks:
Rounding the Lovász Theta Function with a Value Function Approximation, INFORMS, Atlanta, U.S.A. October 26, 2025

Value Function Approximation for Maximum Stable Set Problem, Combinatorial Optimisation and Graph Theory Session, International Symposium on Mathematical Programming (ISMP 2024), Montréal, Canada. July 26, 2024

Value Function Approximation for Maximum Stable Set Problem, First Year PhD Seminar, ISyE, Georgia Institute of Technology. February 23, 2024

Low-Rank Plus Sparse Decompositions of Large-Scale Matrices via Semidefinite Optimization, MMATH Thesis Talk, Department of Combinatorics & Optimization, Faculty of Mathematics, University of Waterloo. May 3, 2023

Convex Optimization, Factor Analysis and Realizability of a Subspace, Graduate Research Seminar, Department of Combinatorics & Optimization, Faculty of Mathematics, University of Waterloo. April 11, 2022

Posters:

Rounding the Lovasz Theta Function with a Value Function Approximation, Mixed Integer Programming Workshop 2024, Poster Finalist, University of Kentucky. March 4, 2024

Deriving Quadratic Assignment Solutions from Semidefinite Relaxations, Mixed Integer Programming Workshop 2026, Poster Finalist, University of Connecticut. May 18, 2026

Workshops:

Attendee: Fulkerson 100, University of Waterloo. July 17-19, 2024.

Algorithms, Combinatorics and Optimization Research Network (ACORN) Meeting 2023, Georgia Institute of Technology. March 9–11, 2023

24th Midwest Optimization Meeting & Workshop on Large Scale Optimization and Applications, University of Waterloo. October 28-29, 2022

WORK

EXPERIENCE

Intact Financial Corporation

Toronto, Canada

Actuarial Analyst of Ontario PL Pricing

2020.01 - 2020.04

Extracted and manipulated databases to respond to internal inquiries by SAS. Generated earning and loss reports weekly and monthly by Excel and SAS. Analyzed correctness of rates for insurance premium algorithm with R.

TEACHING

Teaching Assistant

University of Waterloo

- Undergraduate course: Portfolio Optimization (CO372) 2021.09-2022.04; 2022.09-2022.12

Holding office hours, marking assignments and exams.

- Graduate course: Semidefinite Programming (CO471/671) 2022.05-2022.08

Marking assignments.

TECHNICAL STRENGTHS

Language Proficiency: Native Chinese Speaker, Proficient in English

Programming Languages: Julia (advanced), Python (working proficiency), R (prior industry), C++ (basic), $\text{\LaTeX}$